Product writeup

Carefull transaction alert pipeline · how I approached the problem, what I traded off, and what I would build next

The idea in one paragraph

The data showed that "suspicious" depends on who the person is: a $2,000 Zelle payment is routine for one user and an emergency for another. So the system has two halves. **Cheap, transparent rules** look for the shape of each fraud pattern and are tuned to miss nothing. **An LLM reads each flagged transaction** next to the user's profile and history and decides whether it is genuinely concerning, explains why in words a caregiver can understand, and says how sure it is.

1. What the data told me (notebooks)

- **Three real incidents, two decoys.** Margaret sends growing amounts to a new payee, Dorothy forwards ~94–95% of large deposits from unknown companies within ~30 h, and Helen's card suddenly shows nightlife and luxury hotels in Spain, Thailand and the UAE. Robert (golf, London trips, $2,000 a month to his wife) and Frank (gifts to grandchildren) look unusual but are labeled benign.
- **Size relative to the person separates well.** Benign debits have a median of ~2.5% of monthly income (max ~40%); alerts a median of ~27% (up to ~270%).
- **Generic anomaly detection is noisy.** A 7-day rolling baseline highlighted 78 days, only 17 with a real alert, and still flagged 20 days for Robert and 16 for Frank, who have none.
- **Recipients need cleaning before they can be counted.** The same person appears as `KEVIN R`, `KEVIN RICH` and `KEVIN RICHARDS`; 37 transfers held 13 spellings of 8 people. Without that, "new payee with growing amounts" is invisible.

2. Detection approaches I considered

APPROACH	VERDICT	WHY
Z-score over all of a user's amounts	dropped	Pensions and deposits inflate the mean and deviation; the baseline is not "spending".
Generic rolling-baseline anomaly on daily spend	signal only	Catches spikes but cannot tell a trip from a stolen card; kept as one input, not a detector.
LLM on every transaction	dropped	Pays for 290 calls to confirm groceries, and is harder to audit than explicit rules.
Pattern rules for recall + LLM with context for precision	chosen	**Each rule encodes one fraud shape seen in the data; the LLM supplies the judgment the rules lack.**

PATTERN	RULE (FROM THE NOTEBOOKS)	WHAT THE LLM ADDS
Escalating transfers	Debit > 1.5× the 30-day rolling mean; a second spike within 7 days escalates.	Sees **every transfer to the same (alias-resolved) recipient**, so it can judge the trend and whether the payee is family.
Money mule	Credit followed within 72 h by an outgoing transfer of 85–100% of it.	Knows the sender is a business with no history and the recipient is a stranger.
Out-of-profile spending	Week > 1.5× the 4-week average, only at merchants never used before.	Compares country, category and lifestyle with the biography ("travels frequently" vs. "lives independently, book club").

3. How I handled false positives

159 → 70
rule flags after fixing the weekly rule, with no recall loss

38 → 0
false positives after LLM validation

0
real cases dismissed by the LLM

2 / 2
decoy users (Robert, Frank) left alone

- **Fix false positives at the source first.** My first weekly rule flagged every debit in an expensive week, including the usual grocery store: 146 flags, 122 benign. Limiting it to first-time merchants removed most of them. `check_rules.py` makes this loop fast: change a threshold in `.env`, see true and false positives per rule, no API calls.
- **Give the LLM the person, not just the transaction.** The prompt carries the full profile, the rule evidence with numbers, the user's established behavior and the pattern-specific history. The "established behavior" window stops 30 days before the transaction, so an ongoing incident (Helen's trip) cannot become the baseline that excuses itself.
- **Make it argue both sides.** The output schema asks for evidence *for* and *against* risk before the verdict. That is what lets it say "the $2,000 to Patricia Chen is to his wife, who manages household expenses" instead of reacting to the amount.
- **Never drop uncertainty silently.** Confidence below 0.6 becomes "suspicious + human review"; if the LLM fails, the rule label is kept and reviewed. A missed alert costs more than a review.

4. Tradeoffs I made

DECISION	WHAT I GAINED	WHAT I GAVE UP
Rules tuned for recall, LLM for precision	Every case reaches the LLM; rules stay simple and auditable.	The LLM can only judge what the rules flag: rule recall is the ceiling.
One LLM call per flagged transaction	Simple, per-transaction labels to evaluate and explain.	Helen's 12 purchases cost 12 calls for one story; pattern-level calls would be cheaper.
`gemini-3.1-flash-lite`	~$11/month at 50k transactions, 2.6 s median latency, no errors here.	Not compared yet against a larger model (free-tier quota); untested on harder, ambiguous cases.
Calendar-week totals for the weekly rule	Matches how I explored the data; easy to explain.	Can fire up to 7 days late. Production needs a rolling 7-day window.
Thresholds chosen on the labeled data	Fast iteration with 5 users and 32 positives.	No held-out set: the perfect score is optimistic, not a forecast.

5. Product considerations

- **Escalate gradually, respect autonomy.** The LLM rated Margaret's first $200 payment to Kevin "medium: monitor" and moved to "high: alert" as the amounts grew. In the product, "medium" should mean a gentle check-in with the user, and only "high" should reach the adult children. Alerting on the first payment would feel patronizing; waiting until $2,000 would be too late.
- **The explanation is the product.** Each alert carries a one- or two-sentence summary written for a caregiver ("deposit from an unknown business, almost immediately forwarded to an unfamiliar person") plus the evidence. A caregiver needs to know what to ask, not a risk score.
- **Group alerts by incident.** Dorothy has 12 alerting transactions but one story. Caregivers should get one message per incident, updated as it develops.
- **Profiles are a feature.** The biography is what clears Robert and Frank. Onboarding should collect who the user regularly pays (family, caregivers, contractors) and planned travel.

6. What I would build next

Evaluation

- A held-out set: synthetic users and scenarios generated from the same patterns, plus hard negatives (a user who really does start paying a new contractor more each week).
- Run the comparison with `gemini-3.8-flash` and a non-Google model on those harder cases, where models would actually differ.
- Check whether LLM confidence is calibrated, and tune the review threshold on data with real ambiguity.

System

- Streaming with rolling windows, and pattern-level validation (one call per incident, updated as it grows).
- More patterns: new payee + wire + urgency, gift cards and crypto ATMs, sudden changes in login or contact data.
- Orchestrate the steps with LangGraph, add Batch API and prompt caching, and log prompts and verdicts so caregiver feedback becomes labeled data.